PRODUCT BRIEF / HELMET VERSION 1

Routing Helmet V1

Hear. Understand. Act. Speak. Locally.



Core - the flagship visual direction. Concept render; not fabricated or certified hardware.

THE PRODUCT

A voice-first helmet module that turns short spoken requests into 26 allowlisted actions, keeps core controls available offline, and uses a paired phone for maps, contacts, streaming, and optional AI answers.

Version 1 / September 2026

Pre-prototype product definition and design-to-cost brief

01 / PRODUCT DECISION

A useful helmet computer does not need to know everything

The Version 1 product is deliberately narrow. It listens for a focused request, proposes one typed function call, validates that call against deterministic policy, performs it through approved helmet or phone services, and speaks a short result. Reliability, low distraction, and clear privacy state lead the product; personality makes it enjoyable after the loop works.

ONE-SENTENCE PROMISE

Talk naturally. Get one safe, useful action without touching a screen.

Who it is for

- Daily two-wheeler commuters using navigation, calls, music, and ride capture.
- Touring riders who value offline commands, intercom, and quick status checks.
- Delivery and field fleets that need bounded checklists, notes, and location workflows.
- Helmet and intercom OEMs looking for a testable local-agent module and SDK.

Version 1 commercial shape

DECISION	VERSION 1 POSITION
Beachhead	Phone-paired helmet electronics module or OEM reference design
Core advantage	Offline typed controls with visible privacy and deterministic authority
Design-to-cost	INR 6,000-10,000 recurring module target at volume; not yet quoted
Architecture	Commodity demo hardware first; custom ASIC program remains a measured roadmap

02 / PRODUCT EXPERIENCE

The smoothness is the product

Every command follows one visible, interruptible sequence. The user always knows whether the helmet heard, needs confirmation, completed the action, or safely did nothing.

	STATE	USER EXPERIENCE
01	LISTEN	Low-power wake and voice activity detection. A physical mute remains independent.
02	UNDERSTAND	Short speech recognition produces a transcript and confidence score.
03	PROPOSE	The router emits one closed-schema function and bounded arguments.
04	CHECK	Policy validates allowlist, ranges, motion state, permissions, and confirmation.
05	ACT + SPEAK	The approved service executes; the helmet gives a short result and returns to low power.

Responsiveness targets

MOMENT	TARGET	WHY
Wake acknowledgement	<250 ms	The rider knows the command was heard
Simple local action	<500 ms after speech	Controls feel immediate
Spoken reply begins	<1.0 s target	Conversation feels connected
Physical cancel	<100 ms	The user can always stop output

WHILE MOVING

Replies are short, audio-only, and interruptible. Long answers, list browsing, text entry, and detailed scene interpretation wait until the user is stationary.

03 / VERSION 1 SCOPE

Local first. Phone-backed when useful.

EXECUTION HOME	WHAT IT OWNS
Offline/local	Wake, mute, cancel, short command transcription, typed routing, helmet light and volume control, device-state readback, camera control, ride markers, voice notes, cached navigation cues, and short spoken confirmation.
Paired phone	Calls, contacts, messages, live route start, parked-location lookup, live-location sharing, streaming music services, account setup, OTA, and optional assistant escalation.
Service unavailable	Say what failed, keep the helmet in a safe state, and offer one useful fallback. Never invent successful execution.

Explicit non-goals

- No unconstrained long-form conversation while riding.
- No invisible continuous camera or raw-audio recording.
- No LLM authority over crash decisions, battery protection, thermal shutdown, or vehicle control.
- No claim that a modified certified helmet automatically remains certified.
- No promise that three custom ASIC packages are the lowest-cost production architecture.



Urban - a calmer commuter form with a serviceable rear electronics zone. Concept only.

04 / CAPABILITY MAP

Eight families. One voice surface.

01 Camera Take a photo, start/stop video, mark a ride moment.	02 Music Play, pause, skip, check the track, and set safe volume.
03 Lights Control visibility light and brightness through policy limits.	04 Navigation Start saved routes, repeat cues, and locate the parked vehicle.
05 Calls Answer, reject, call a known contact, and send quick messages.	06 Notes Record an indicated voice note and share location for a fixed time.
07 Helmet state Read battery, connectivity, temperature, and fault summaries.	08 Emergency Hand an SOS request to a separate deterministic controller.
PROVIDER ABSTRACTION **The rider says 'play music' or 'call Alex.' The helmet calls one stable media or phone function. Spotify, Apple Music, local playback, contacts, or maps sit behind the phone service and can change without changing the voice contract.**	

Feedback language

- Acknowledge quickly: 'Got it.'
- Confirm only when needed: 'Start recording video?'
- Report the actual result: 'Recording started.'
- Fail plainly: 'Your phone is not connected. The helmet light still works.'

05 / FROZEN FUNCTION CATALOG

Local controls, status, media, and intercom

These are the 26 currently frozen proposals. 'Local' means the proposal can complete without an internet service when the associated hardware is present; it does not mean the language model executes the action directly.

FUNCTION	EXAMPLE PHRASE	RUN	POLICY
set_helmet_light(on, brightness)	Turn the helmet light on at forty percent.	Local	Immediate
read_battery()	How much battery is left?	Local	Read only
read_device_health()	Run a helmet check.	Local	Read only
read_connectivity()	Is my phone connected?	Local	Read only
set_audio_volume(percent)	Set the volume to thirty percent.	Local	Immediate
mute_audio(muted)	Mute the helmet.	Local	Immediate
set_music_playback(action)	Play the next song.	Phone	Immediate
get_music_status()	What song is playing?	Phone	Read only
set_intercom_mode(action, group_id)	Join my riding group.	Local	Confirm

TYPED, NOT IMPROVISED

Unknown function names, extra arguments, out-of-range values, low confidence, stale requests, and unauthorized calls fail closed.

05 / FROZEN FUNCTION CATALOG

Camera and navigation

These are the 26 currently frozen proposals. 'Local' means the proposal can complete without an internet service when the associated hardware is present; it does not mean the language model executes the action directly.

FUNCTION	EXAMPLE PHRASE	RUN	POLICY
set_camera_recording(recording)	Start recording video.	Local	Confirm
take_photo()	Take a picture.	Local	Confirm
save_event_marker(label)	Mark that moment.	Local	Immediate
repeat_navigation_instruction()	Repeat the last direction.	Local	Read only
navigate_to_saved_place(place_id)	Navigate home.	Phone	Confirm
locate_parked_vehicle()	Where did I park?	Phone	Read only

RECORDING BOUNDARY

Camera and voice-note actions always expose a hardware-backed visible state. One spoken stop command and the physical cancel control remain available.

05 / FROZEN FUNCTION CATALOG

Calls, messages, notes, sharing, and emergency control

These are the 26 currently frozen proposals. 'Local' means the proposal can complete without an internet service when the associated hardware is present; it does not mean the language model executes the action directly.

FUNCTION	EXAMPLE PHRASE	RUN	POLICY
answer_call()	Answer the call.	Phone	Confirm
reject_call()	Reject the call.	Phone	Immediate
call_contact(contact_id)	Call Alex.	Phone	Confirm
send_quick_message(contact_id, template_id)	Tell Alex I am running late.	Phone	Confirm
read_last_message()	Read my latest message.	Phone	Confirm
start_voice_note(label)	Start a voice note.	Local	Confirm
stop_voice_note()	Stop the voice note.	Local	Immediate
share_live_location(contact_id, minutes)	Share my location with Alex for thirty minutes.	Phone	Confirm
set_do_not_disturb(enabled)	Turn on do not disturb.	Local	Immediate
request_emergency_call(reason_code)	Call emergency services.	Phone	Emergency path
cancel_pending_action()	Cancel that.	Local	Immediate

EMERGENCY BOUNDARY

The assistant can request an emergency flow. An independent deterministic controller owns countdown, cancellation, calling, and location dispatch.

06 / DEMONSTRATION GROWTH

Add capability without widening authority

The following calls are product candidates after the frozen V1 catalog. They should enter the demo one by one, each with a closed schema, privacy behavior, motion policy, failure path, and policy-corpus tests.

PROPOSED CALL	PRODUCT MEANING
ask_assistant(query, response_mode)	Ask a general question; phone/cloud-backed with short ride-safe replies.
ask_assistant_with_camera(question)	Use one user-triggered frame as context; explicit camera indication.
play_media_query(query, provider)	Resolve a song, artist, album, or playlist through the phone.
identify_song()	Identify nearby music through a phone service.
describe_scene(detail_level)	Describe a user-triggered image while stationary.
read_visible_text(language)	Read or translate a sign from one captured frame.
find_nearby(category)	Find fuel, food, service, parking, or medical help.
set_headlamp(on, brightness, beam)	Optional dedicated forehead lamp with deterministic limits.
create_ride_recap()	Summarize route markers, photos, and notes after the ride.

ASSISTANT QUERIES

General questions are a phone/cloud tool, not unrestricted device authority. The response is summarized for speech, and long answers are deferred while moving.

07 / PHYSICAL PRODUCT

A serviceable module around a certified protective product



Response - electronics remain visually separate from the protective shell. Concept render only.

ZONE	VERSION 1 REQUIREMENT
External pod	No drilling or cutting the shell/EPS in the engineering demonstrator; sealed, removable, and inspectable.
Audio	Wind-protected microphones, helmet-pocket speakers or open-ear transducers, AEC/noise handling, independent limiter.
Controls	Large physical mute/cancel; smaller action/pair button; glove-operable feedback.
Indicators	Hardware-backed recording, connection, charging, and fault state.
Power	Protected battery, thermal sensing, safe charging, service plan, and independent shutdown.
Phone	Bluetooth/audio transport, maps, contacts, services, setup, signed OTA, and optional cloud escalation.

08 / COMPANION FRONTEND

One glance before the ride. No screen work during it.

The companion app is for setup, permissions, saved places, contacts, service providers, updates, and post-ride review. During motion, the helmet uses voice, tones, a physical cancel control, and concise status lights.

HOME	RIDES	TOOLS	PRIVACY
Connected Battery 82% Recording off Light 40%	Markers Photos Notes Ride recap	Saved places Quick contacts Music provider Intercom groups	Mic state Camera state Cloud policy Delete local data

Voice states

STATE	SOUND / LIGHT	SPOKEN BEHAVIOR
Ready	Short wake tone	No unnecessary chatter
Needs confirmation	Amber pulse	Repeat action and one critical argument
Recording	Persistent visible indicator	Start/stop acknowledgement
Phone unavailable	Connection tone	Name the unavailable service and local fallback
Fault / over-temperature	Distinct fault pattern	Stop nonessential work; deterministic protection owns response

DESIGN RULE

The app may be rich. The ride experience stays sparse.

09 / TRUST BOUNDARY

The model proposes. Deterministic systems decide.

LAYER	CAN DO	CANNOT DO
Speech/router/voice models	Transcribe, propose a known function, generate a short reply	Write actuator bytes, bypass confirmation, or create a privileged tool
Policy supervisor	Validate schema, confidence, state, ranges, permissions, and motion policy	Redefine helmet certification or ignore hardwired safety inputs
Emergency controller	Own SOS countdown, cancel, calling, and location flow	Depend on the conversational model to remain safe
Power/audio safety	Own battery protection, thermal shutdown, volume limit, and physical mute	Be overridden by an assistant response

Privacy is a physical feature

- Raw audio remains local by default and is not stored without explicit policy.
- Recording state has a hardware-backed indicator; camera use is user-triggered.
- Cloud escalation is explicit, visible, and bounded by user preference.
- Firmware and model artifacts are signed, versioned, and recoverable through rollback.
- Logs minimize raw voice and expose delete/export controls in the companion app.

CERTIFICATION BOUNDARY

An integrated sellable helmet must be co-designed and tested with a helmet OEM. The concept does not assume that adding electronics to an already certified shell preserves certification.

10 / DESIGN TO COST

Target the module first: INR 6,000-10,000

This is a planning model for 1,000-10,000 unit volume, not a supplier quote. The design center is approximately INR 8,200 for the full camera/audio module; a camera-free Lite configuration can target approximately INR 5,500-6,500.

RECURRING MODULE BLOCK	INR / UNIT
Compute, control, Bluetooth/audio radio, memory	1,300-2,400
Camera and storage interface	450-950
Microphones, codec/amplifier, speakers	550-1,050
Battery, charger, protection, power conversion	550-1,000
Lights, controls, indicators, basic sensors	250-550
PCB/FPC, antenna, connectors, harness	450-850
External pod, seals, mounts, acoustic parts	400-800
Assembly, test, and yield reserve	600-1,200
Electronics/module subtotal	**4,550-8,800**
Inbound logistics and initial warranty reserve	400-900
Planning landed range	**4,950-9,700**

EXCLUDED FROM THE RANGE

Helmet shell, custom-silicon NRE, certification, one-off prototype pricing, GST, channel margin, app/cloud operations, liability, and full service infrastructure. A sub-INR 10,000 consumer retail promise needs a much lower landed cost and supplier quotes.

11 / BUILD SEQUENCE

Grow the demo by earning each layer

STAGE	DELIVERABLE	PASS CONDITION
01 Software loop	Speech -> typed proposal -> policy -> mock action -> spoken reply	26-call catalog and reject cases run reproducibly
02 Bench module	Commodity compute, audio, buttons, lights, camera, battery, phone bridge	Core actions work with visible state and safe failure
03 Helmet fixture	Non-structural removable pod and real microphone/speaker placement	Noisy-road intent and cancel tests pass
04 Pilot app	Setup, permissions, contacts, saved places, providers, OTA, logs	Small rider pilot completes target tasks without screen interaction
05 OEM pilot	Co-designed mechanics, acoustic, RF, battery, service, and certification plan	Written OEM path plus supplier-quoted COGS
06 Silicon decision	Compare commodity SoC, one shared accelerator, and three research ASICs	Custom silicon wins a measured customer-valued metric

Version 1 acceptance gates

- No unauthorized or malformed proposal causes physical action.
- Every recording state is visible and every output is physically cancellable.
- Core local controls work with the phone disconnected.
- The system reports real completion/failure rather than a conversational guess.
- Noisy-road accuracy, commute-day power, thermal, and supplier cost are measured before claims expand.

NEXT BUILD

Ship the commodity-hardware bench module and companion setup screen first. Keep the 26-call contract stable while camera, media search, and assistant extensions are added behind policy.